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Finding Plankton: A Marine Science Scavenger Hunt for Local Communities

Hannah M. Budroe, and Holly M. Bik

Abstract

At a time of increasing political polarization and rapidly accelerating climate change, it is important to build public knowledge and empathy toward nature to foster behavioral changes; however, addressing a knowledge deficit can be insufficient to affect these changes. To leverage the intersection of science, art, and intergenerational learning to move toward such outcomes, we describe here a multi-part public outreach initiative. Briefly, this initiative was designed to build community knowledge and empathy with the ocean while encouraging curiosity and enjoyment in science learning. This involved interactive tabling at the local farmers market, as well as a marine biology-themed scavenger hunt through which participants collected plankton-themed trading cards from local businesses. Through the scavenger hunt, families learned about various plankton groups in coastal Georgia, resulting in approximately 256 participating individuals in one month. The activity was accessible across age groups, gamified science learning, and enhanced intergenerational learning. We believe such activities could be more broadly applicable, and so also provide recommendations for orchestrating similar science communication campaigns. Such marine-focused education is particularly important in landlocked cities where interactions with the ocean may be limited, and can hopefully instill an ongoing interest in marine sciences and empathy for ocean conservation efforts.

Introduction

At a time of increasing political polarization and rapidly accelerating climate change, it is important to build public knowledge and empathy toward nature to foster behavioral changes. Most science research is primarily shared as "inreach" (Bik et al. 2015) to the already established scientific community in the form of scientific papers or conference presentations, thus creating a barrier to the public through inaccessible language and paywalls. Existing scientific communication efforts can be effective but often run into pitfalls such as assuming the only obstacle to changing public beliefs and behavior is fixing a knowledge deficit on the subject (Besley and Dudo 2022). Many science communication researchers have shown insufficient knowledge is not the only piece of the puzzle to changing public perception—building trust and reframing the narrative are also important (Besley and Dudo 2022). Because of these challenges, new creative and multigenerational approaches to science communication may be an effective way to overcome existing barriers in public outreach aimed to result in behavioral change.

Child-to-parent intergenerational learning may be one method through which educators can facilitate paradigm shifts in adult opinions of climate change (Lawson et al. 2019). Generally, adolescents are more receptive to climate change education than adults, with their opinions less influenced by socio-ideological factors. Children can pass on their own concerns and ideas to their parents, who might otherwise be less receptive to these ideas if conveyed from an outside party than from their own children. Thus, "bottom up" science campaigns with a focus on children may be an effective way to disseminate important climate messages across families rather than a "top down" approach focusing on educating parents. However, effective science outreach to children also comes with its own hurdles, requiring engaging and memorable content for children. Children may struggle with activities requiring significant attention spans or memorization. Instead, children are more likely to remember scientific content through forming emotional connections and active participation in experiences, while those only observing an activity may be less likely to retain that information (Idema and Patrick 2019). In particular, interactive activities and visual art can be used to sustain engagement and information recall in children.

Visual art can be effective in translating complex concepts while also strongly sustaining audience engagement and information recall (Kaufmann et al. 2023). For example, when interacting with a science art exhibit about water quality, audience members experienced greater emotional engagement at the time and greater information recall several months after the event compared to those only reading the same information in a static booklet. These types of active learning through art, as opposed to passively reading or watching a video on the same subject, can enhance an audience's understanding and attention to a subject. Visual art can also help build empathy with nature (Curtis 2009)—a necessary emotion to evoke behavioral change.

To leverage the intersection of science, intergenerational learning, and visual art, we describe here a multi-part public outreach initiative to build community knowledge and empathy with the ocean in an inland city while encouraging curiosity and enjoyment in science learning.

Our plankton scavenger hunt

In the city of Athens, Georgia, USA surrounding the University of Georgia (UGA),

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we built a marine biology-themed scavenger hunt through which children (and surprisingly many adults) collected plankton-themed trading cards (*see* Fig. 1). Athens is a socioeconomically diverse city with a population of around 128,000 located 4–5 hours away from any coastline, but with a research-heavy Marine Sciences department located on the UGA campus. Due to this geographic location, ocean education is complicated by distance and financial barriers which may prohibit a large subset of the local population from experiencing ocean habitats. Most likely, many children have had limited exposure to marine biology and prior knowledge is primarily of larger charismatic megafauna. The Georgia Aquarium located in Atlanta, approximately 2 hours from Athens, has many exhibits of larger fish and sharks, but not many planktonic organisms beyond a few jellyfish. Additionally, while the aquarium is significantly closer than Georgia's coast, the transportation and cost of entrance are not insignificant for families. Trips to the beach are even further and more logistically complicated, and thus many locals may go most of their childhood without experiencing marine life. Because of this, a low-cost, interactive activity that can effectively entertain and educate these communities about the ocean is important.

Trading card design

As part of two NSF grants (to Bik) and an institutional science communication grant (to Budroe), we designed an educational deck of eight "taxonomy trading cards" representing various plankton species to enhance knowledge about biodiversity in the ocean (Fig. 1). Information on the cards includes broad taxonomy, a hand-drawn image, general characteristics, and the organism's ecological role. Trading cards were modeled after popular card games and trading decks, including colorful icons to describe the plankton "type" (e.g., grass for phytoplankton, wind for *Velella velella*, etc.) and a weakness section describing predators and environmental threats (e.g., ocean acidification, etc.). Facts were scientifically accurate for the species being depicted. Organisms included common phytoplankton and zooplankton off the Georgia coast in addition to global plankton species with unique ecological roles or interesting stories. To spark local interest, we included several species that are particularly relevant to Georgia coastal ecosystems, including doliolids (pelagic tunicates) that frequently bloom in the South Atlantic Bight, and dinoflagellates causing harmful algal blooms (HABs) that are currently being studied by researchers at UGA and Skidaway Institute of Oceanography. Trading cards can be downloaded for use via Budroe's GitHub repository (https://github.com/hbudroe/ScienceCommunication.git).

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We also designed a stamp card for participants to track their progress across the scavenger hunt. This tracking card included the names of the eight stations next to a space for stamps to mark off each location. The back also included a short blurb introducing plankton and a QR code leading to additional information on the Plankton Chronicles website (www.planktonchronicles.org).

The scavenger hunt

During the scavenger hunt, participants visited eight local institutions to collect a unique trading card and learn about the plankton depicted on the card. At each station, participants collected a unique trading card and stamped off the corresponding location on their tracking card. Purchases were not required to participate. Businesses included an indie bookstore, plant store, the children's desk of the public library, a kombucha brewery, a children's craft and toy store, and a board game cafe. Stations were initially chosen based on criteria such as independently owned, family friendly, within a close radius of downtown Athens, and visited by a slightly different demographic than other locations. We then visited locations to assess suitability and inquire with staff in person. Formal emails were sent explaining the purpose of the scavenger hunt, its duration, details, and materials that would be provided. Responses were positive, with all owners excited to participate. The final "scavenger hunt kit" provided to each station included: a descriptive flyer, a specific trading card, a stamp card listing each station, marine-themed ink stamps for participants to mark each station, and a QR code to collect survey feedback (*see* Fig. 2b–d). The range of stations allowed the cards to reach people of a broad array of ages and backgrounds while also encouraging them to explore new parts of their city. Scavenger hunt materials were displayed on the front counter of each business, catching customers' attention during checkout. Employees at the children's craft store even decorated their display with colorful drawings of the

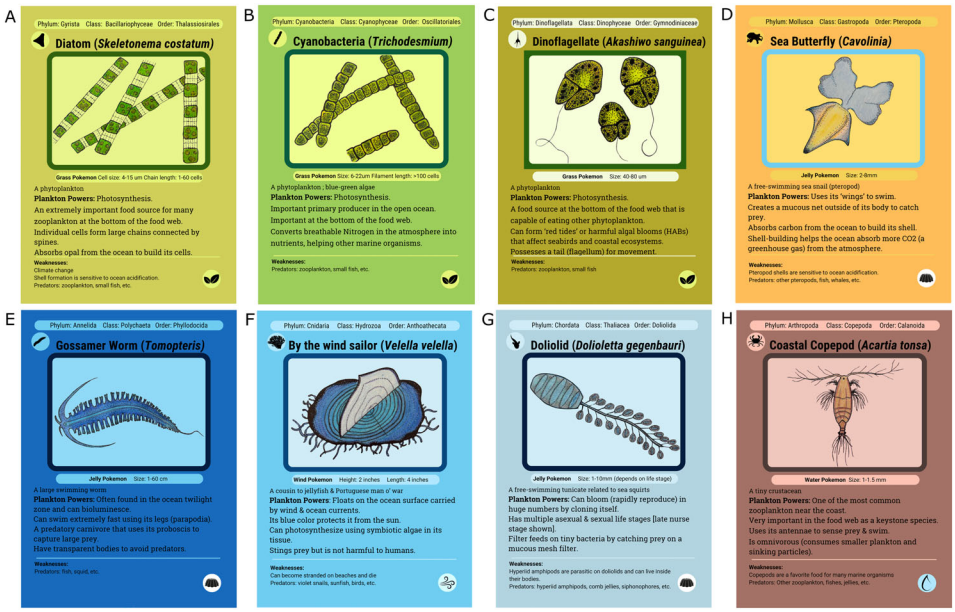


FIG. 1. Distributed trading cards depicting cosmopolitan or Georgia-specific coastal (**a–c**) phytoplankton and (**d–h**) zooplankton.

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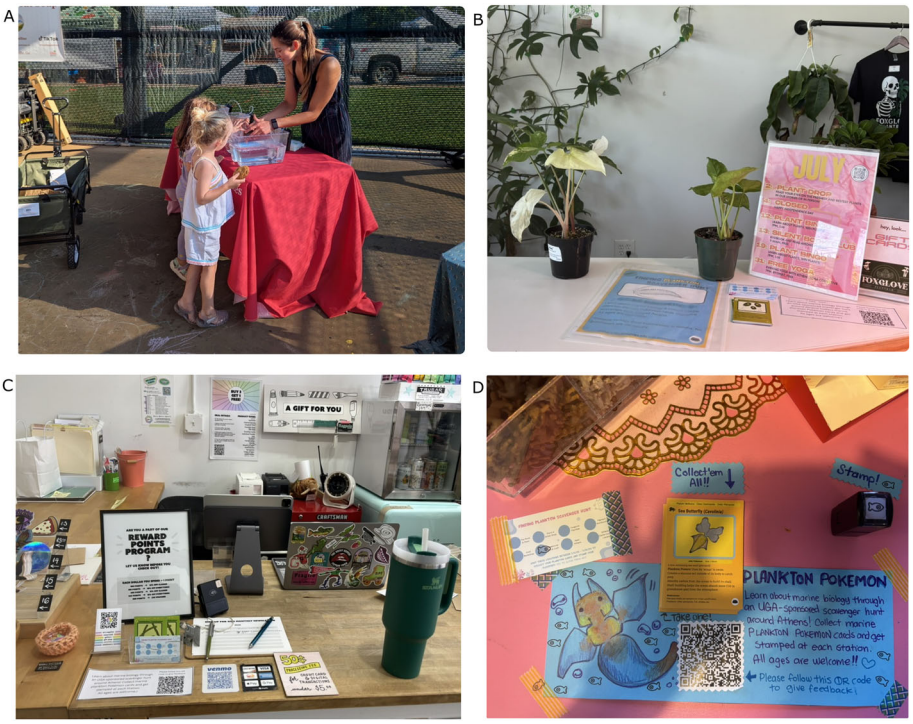


FIG. 2. Physical outreach and advertising for the scavenger hunt at select stations included (**a**) Budroe tabling at the farmers market's children's table and introducing children to live invertebrates. Also included are pictures of Finding Plankton displays at (**b**) Foxglove Plantbar, (**c**) K.A. Artist Shop, and (**d**) Treehouse Kid & Craft.

pteropods from their plankton card to further attract attention (Fig. 2d).

To advertise the scavenger hunt and further build community interest in marine science, we also organized an outreach event at the local farmers market modeled as a miniature science festival. The Athens Farmers Market is held weekly on Saturday mornings at a park, hosting more than 20 vendors and drawing a large attendance from the Athens community. In addition to the normal artisan and farmer stands, the market also boasts a "children's table" that changes weekly and offers activities to entertain children while their parents peruse the market. In previous weeks this has included a mini-library, music and crafts, and other activities. Budroe took over the children's booth with several graduate student volunteers, bringing a touch tank of live invertebrates as outreach animals (Fig. 2a), as well as ocean-themed crafts, and the scavenger hunt starter pack (a tracking card and one of the trading cards) to garner initial interest. Both children and adults were excited to meet the animals and learn about the scavenger hunt. Many of the children had never seen marine invertebrates before. We were asked many questions by adults and children about how we became scientists, what it is like to perform fieldwork, and other day-in-the-life questions. Several parents encouraged their children to talk to us, mentioning that their child was interested in science or dreamed of becoming a marine biologist. These face-to-face interactions helped children become more familiar with local coastal invertebrates, allowed parents to learn more about research being performed locally, and provided representation of real-life scientists. Several of us graduate students were from underrepresented populations in STEM (e.g., in terms of gender and racially minoritized backgrounds), and seeing real scientists that may look different from those represented in media may have served as additional encouragement to younger children (Jin 2024). This outreach made a step to bridge the gap between public perception of academic research and its reality.

Monitoring success

To monitor the scavenger hunt's success, we tracked several metrics (*see* Table 1). We measured the number of stamp cards and trading cards collected by participants from each station, and tracked interactions at the farmers market. We also visited the stations throughout the month to monitor progress and restock materials as needed. The scavenger hunt ran for the entire month of July 2025 and resulted in at least 256 individuals participating at least once. The children's toy and craft store was the most popular station and frequently ran out of stamp tracking cards, followed by the board

TABLE 1. Participation metrics and qualitative feedback from the scavenger hunt

Total participants	Stamp cards distributed at farmers market	Total plankton cards collected	Average # plankton cards collected per person
256	54	897	3.5

Qualitative feedback from participant surveys, farmers market outreach, and stations:

- "*Super excited to start*!"
- "*This looks so fun! My daughter loves Pokémon cards & the ocean so she'll love this*!"
- "*This was fun*!!! *They are beautiful*."
- "*They look very nice. Thick cardstock. And an admirable Pokémon tribute*."
- "*The design at the Treehouse was so cool*!"
- "*Very informative. Love the weaknesses part*."
- "*Customers loved the scavenger hunt. We had a lot of people come in looking for it*."

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BOX 1. Lessons learned and recommendations for future community-based outreach programs.

1. ***Have a good team of people***. A good team of people can be helpful in implementing these ambitious events. While we were expecting to dedicate several hours creating materials, we had not anticipated the additional time cost of transporting and collecting materials to and from each scavenger hunt station. Additionally, the scavenger hunt was more popular than originally expected, resulting in the need to print and transport additional materials for distribution. A team of individuals to help with material distribution and monitoring progress would be useful in implementing a smooth program.
2. ***Budget carefully (and generously, if possible)***. $1000 USD was allocated to Budroe through a communication grant, the majority of which was spent printing high-quality printing cards. 500 cards were printed for each trading card design (500 cards × 8 designs = 4000 total cards) for approximately $600, while the remaining funds were spent in printing the stamp cards and purchasing marine-themed stamps for each station. If it was available, additional budget could be used to hire a graphic designer or social media coordinator to advertise the event and aid in designing materials. Additional funds could also be used to print the trading cards in higher quality with thicker paper and a glossier finish.
3. ***Consider the timing of events, and how long they should run***. An event like this could run for multiple months, giving many individuals the opportunity to participate and spread the word. We ran this scavenger hunt for one month based upon a timeline given by the deadlines for the grant Budroe received. Upon ending the scavenger hunt, many individuals still had not completed the activity due to summer travel and some even commented in the survey they wished it had lasted longer. If organizing a similar program in the future, it could be useful to run the event for the entire length of summer or coinciding with the school year in some way to maximize participation.
4. ***Have extensive advertising***. Budroe advertised the scavenger hunt primarily through tabling at the farmers market and flyers at the stations themselves. More advertising through additional flyers and social media posts, paired with advertisements in online community calendars and cross-posting on the business' social media feeds, could reach broader audiences. If the scavenger hunt overlaps with the K–12 school year, direct communication with local teachers could also encourage additional participation by their classes.
5. ***Collect a lot of data***. The survey linked to the QR code collected some useful data, including answers for where participants first learned about the scavenger hunt, their age, the number of cards collected, comments about the cards and the activity, and their favorite card. Further demographic data could be useful, particularly to distinguish between adults answering the survey for their child versus on their own. Moreover, it would be useful to collect additional information, including a survey of people's knowledge or attitude changes toward marine plankton and ocean conservation. More formal questions assessing knowledge gained or changes in behavior directly linked to the activity would be extremely useful in evaluating the educational efficacy of the trading cards. For example, the UGA Center for Teaching and Learning offers assistance from assessment specialists—trained social scientists—that may be able to design better surveys and indicators of learning outcomes. Furthermore, the survey received a low response rate—alternative methods for garnering more attention could be including the survey QR code on the stamp card alongside a note to complete when finished with the activity, verbal questions by an organizer to participants at one of the more popular stations during high activity hours, or a stack of printed questionnaires and collection box next to the displays.
6. ***Make it free to participate***. While children are on summer vacation, many parents may prefer free activities to engage their children as an alternative to expensive summer camps or other costly activities. That our activity could be done without required purchases allowed families to visit each business and peruse without feeling the obligation to buy any items. In general, free activities are much more likely to engage people of all ages, as even the idea of having to spend money or pull out one's wallet can create a mental barrier to participation. No- or low-cost activities are also likely to reach a more socioeconomically diverse group of participants, especially if the local businesses cover different geographic areas and neighborhoods.
7. ***Have hands-on activities***. Physical outreach products grab people's attention even when they are not looking to participate. It also helps to reach those without internet access due to age or other circumstances, and many people enjoy collecting physical objects as opposed to digital images. Participants also mentioned enjoying the tactile feel of the cards themselves. At the farmers market, children really took part in the touch tank and enjoyed the different sensations of each animal. Even adults still appreciated the opportunity to touch something "slimey" or just different, which is more memorable than reading about the same animal. The touch tank events provided additional outreach opportunities at the farmers market even if participants did not ultimately take a trading card or participate in the plankton scavenger hunt.
8. ***Incorporate multiple levels of engagement***. Different levels of the outreach activity may appeal to various audiences. Smaller children unable to read still enjoyed the artistic appearance of the card, overall resemblance to Pokémon cards, and cute fish stamps received at each station. Children particularly enjoyed being able to mark off stations with the stamps. Some participants, especially science graduate students, were more interested in the educational content of the cards, while others were initially drawn in by the artistic appearance. Many adults also appreciated the Pokémon resemblance, appealing to their sense of nostalgia and whimsy. For later recall, people may not remember all of the facts on the cards, but might still recognize the general appearance of the plankton. For those that may not have the bandwidth to engage with all eight of the stations, even visiting one station or the farmers market still served as a complete educational experience.

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game cafe and the plant store. Based on the distribution ratio of stamp cards to trading cards, participants averaged collecting approximately 3.5 plankton cards per person. Qualitative feedback was also collected through an online survey and conversations with front desk staff at each station, as well as conversations with families at the farmers market. Of those completing the survey, around 75% were over the age of 20, although this is potentially biased toward adults possessing smartphones with a capability to scan QR codes. Feedback in the survey was positive, with participants admiring the card designs and drawings, facts about marine species, and the overall scavenger hunt experience. Participating businesses were excited to take part for the community involvement and built-in advertising. At the end of the month, most staff raved that people had loved the cards and that the scavenger hunt was extremely popular. Many participating businesses noted that during the scavenger hunt they received many new customers (although purchases were not required for collection) or individuals who had never heard of their business before. Additionally, while it was expected that children would enjoy the plankton cards, a surprising number of adults seemed even more excited than the children. For example, we distributed flyers and tracking cards around several academic buildings on UGA campus, resulting in participation by many graduate students. Parents were also exposed to the cards and the general concept of plankton through facilitating their children's participation (transporting them to each station and helping keep track of materials), in addition to any intergenerational learning that may have occurred through conversations with their kids.

Conclusions

This scavenger hunt served as an exercise that educated children and parents about the ocean, instilled a sense of empathy with marine life, and connected participants with their local community. The trading cards were accessible across age groups and gamified science learning, helping to teach individuals about the existence of plankton and strengthening their connection to the ocean. Education occurred at multiple levels. The cards provided a recognizable face of marine plankton through the common species name and species drawing—for example, individuals may now recognize the concept of "plankton" or the word "pteropod" if heard again. Having a "face" to these organisms can also help build a sense of empathy and connection with animals that might have previously appeared distant. Those more interested in the content of the cards might recall some of the facts for each card or also might begin to build an interest in these "cool" organisms. Furthermore, the fun card design and game aspect of the collection process, akin to "Pokémon Go," could help build a stronger positive association with science learning as opposed to previous classroom lessons that may be remembered as more boring. Finally, individuals may be more likely to keep these cards given their fun appearance and an implicit habit of collecting Pokémon cards. This strengthens science learning and positive associations, supporting future interest in ocean science. Because of this excitement about the cards, children may also be more likely to share that excitement and information gained with their parents, friends, or other family, thereby enhancing cross-individual and intergenerational learning. In creating and implementing the activities described here, we learned a number of lessons that could be useful for anyone undertaking future community-based outreach programs (summarized in Box 1). While such marine-focused education is likely of value everywhere, we believe it is particularly important in landlocked cities such as ours where interactions with the ocean are limited and restricted primarily to charismatic megafauna. Interacting with ocean biodiversity through the plankton cards will hopefully instill an ongoing interest in marine sciences and empathy for ocean conservation efforts.

Acknowledgments

Funding for execution was provided by the University of Georgia Graduate School. Funding for this work was provided by a National Science Foundation CAREER award (DEB-2144304) and NSF Antarctic Program award (OPP-2132641) to Holly M. Bik at UGA. We would like to give a huge thank you to graduate students in the UGA Marine Science Department, including Abby Swierz, Alejandro De Santiago, Claire Garfield, Michelle DeMers, and (honorary graduate student) Kaitlyn Oberg for their help in advertising, scavenger hunt set-up, and tabling at the farmer's market. Thank you to Dr. Austin Heil and Angelica Dziurzynski for loaning the teaching lab outreach animals for the farmers market event. An additional thank you to ComSciCon Atlanta 2025 for hosting the Create-A-Thon that prompted the original conceptualization of the trading cards and provided expert feedback on the designs.

Author contributions

Hannah M. Budroe designed the cards and scavenger hunt. Holly M. Bik contributed to the scavenger hunt design and helped revise the trading card design. Hannah M. Budroe secured the execution funding and managed the physical outreach for the events. Hannah M. Budroe wrote the paper and Holly M. Bik contributed to editing.

Conflicts of interest

The authors have no conflicts of interest to disclose.

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